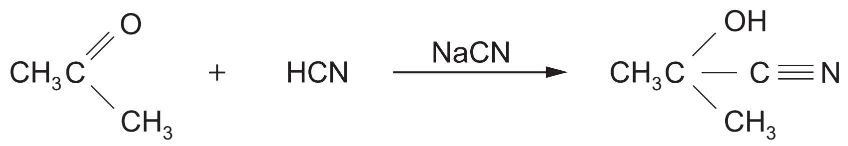
Examiner only

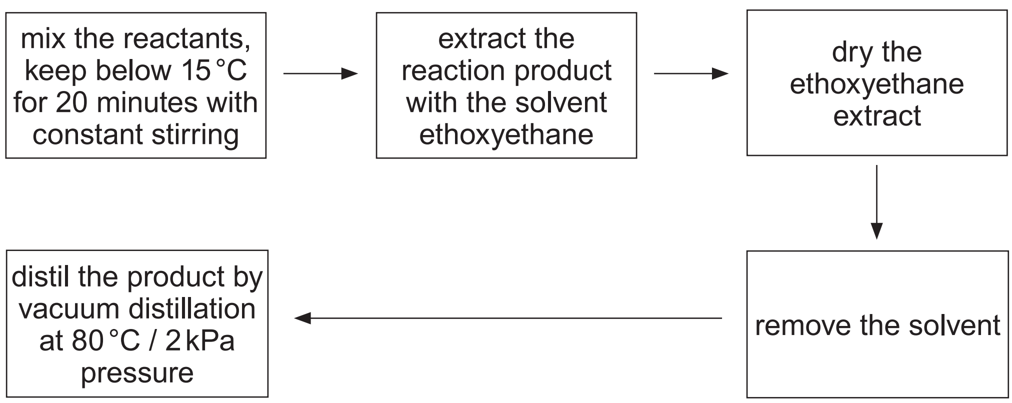
11. *(a)* Propanone reacts with hydrogen cyanide (in the presence of sodium cyanide) to produce 2-hydroxy-2-methylpropanenitrile.



(i) Draw the mechanism for this reaction. Name the type of reaction mechanism occurring. [3]

Mechanism type ....

(ii) The following procedure was suggested to obtain 2-hydroxy-2-methylpropanenitrile.



The overall percentage yield was 55 %.

Suggest **two** stages in the procedure of extraction and distillation that could have resulted in this reduced yield. [2]

1. ....

....

2. ....

....

(iii) In a further experiment an excess of hydrogen cyanide reacted with 17.4 g of propanone (M_r 58.06) to produce 18.6 g of 2-hydroxy-2-methylpropanenitrile (M_r 85.07).

Calculate the percentage yield of 2-hydroxy-2-methylpropanenitrile. [3]

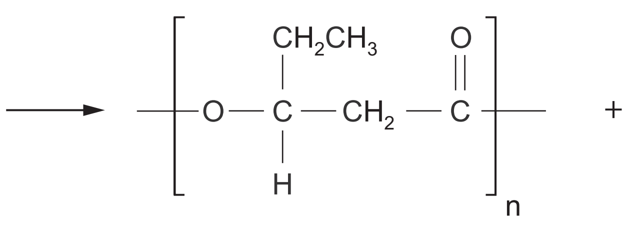
Percentage yield = ... %

(iv) The dehydration of 2-hydroxy-2-methylpropanenitrile produces 2-methylpropenenitrile, $H_2C{=}C(CH_3)CN$. This compound can undergo addition polymerisation giving poly(2-methylpropenenitrile).

Write the formula of the repeating section of this polymer. [1]

(b) There is considerable interest in biodegradable polymers. One of these is the polyester 'polyhydroxyvalerate' (PHV), which is produced from starch or glucose by using microorganisms. One simple chemical way of producing PHV is by the condensation polymerisation of 3-hydroxypentanoic acid.

Complete the equation below, which shows the structure of the repeating polymeric unit. [2]



3-hydroxypentanoic acid

Examiner only

(c) State **one** difference between condensation polymerisation and addition polymerisation. [1]

....

....

(d) Alcohols react with sodium to give hydrogen gas as one of the products.

$$2R—OH \quad + \quad 2Na \longrightarrow 2R—O^{-}Na^{+} \quad + \quad H_2$$

A student thought that the volume of hydrogen given off could be used to identify the alcohol.

In an experiment 0.900 g of the alcohol were dissolved in an inert solvent and an excess of sodium metal was added. 184 cm^3 of hydrogen were collected, measured at 298 K and 1 atm pressure.

(i) The reaction is exothermic. Suggest how the reaction mixture could be maintained at close to room temperature. [1]

....

....

(ii) Use the figures to calculate the relative molecular mass of the alcohol. [2]

M_r = ...

(iii) The alcohol used for this experiment gave a ketone on oxidation.

Give the displayed formula of the alcohol. [1]

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